Lecture 5

Domain of dependence

Lecture 4 constructs a field from data on all of space. The local energy balance gives a stronger conclusion: data far enough from a point cannot change the field at that point. We prove this for the real Klein–Gordon field with $m \geq 0$ and propagation speed one.

Objectives. *We prove an energy inequality on shrinking balls and deduce local uniqueness and finite propagation speed. A compact initial pulse and the heat equation show what the propagation statement does and does not say.*

By the end of this lecture you will be able to:

- Derive the nonnegative energy flux through a backward cone.
- Identify the initial region determining a field at a point.
- Distinguish finite propagation speed from Huygens' principle.

5.1 The region determining a point

Write $B_r(a) = \{x \in \mathbb{R}^d : |x - a| < r\}$, where $d \geq 1$. We prescribe data at $t = 0$. A light cone has boundary slope corresponding to speed one.

Definition 5.1 (Dependence and influence regions)**.** For an event (T, a) with $T > 0$, its *backward cone* is

$$C^-_{T,a} = \{(t, x) : 0 \leq t \leq T,\ |x - a| \leq T - t\}.$$

An initial region is a *domain of dependence* for (T, a) if any two C^2 Klein–Gordon solutions with equal data there have equal values at (T, a). For a closed initial set $K \subset \mathbb{R}^d$, its *range of influence* is

$$J^+(K) = \{(t, x) : t \geq 0,\ \operatorname{dist}(x, K) \leq t\}.$$

The energy argument below proves that the base $\overline{B_T(a)}$ is a domain of dependence for Klein–Gordon solutions. Figure 5.1 shows the shrinking slices of one backward cone. Each boundary line has speed one.

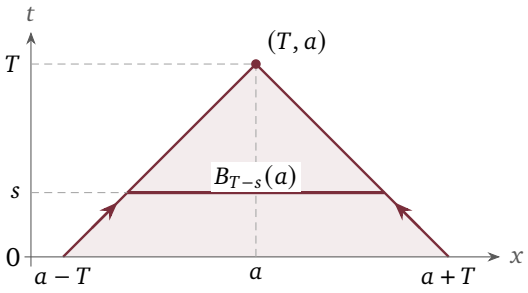


Figure 5.1: A spacetime section of $C^-_{T,a}$ through one spatial axis. The slice at time s is the ball of radius $T - s$. Its lateral boundary moves inward at speed one.

For $u \in C^2$ satisfying $(\Box + m^2)u = 0$, Lecture 3 gives the C^1 energy density and spatial energy flux

$$\rho_{\mathrm{en}} = \frac{1}{2}\bigl(u_t^2 + |\nabla u|^2 + m^2 u^2\bigr), \qquad \mathbf{f} = -u_t \nabla u, \qquad \partial_t \rho_{\mathrm{en}} + \operatorname{div} \mathbf{f} = 0. \tag{5.1}$$

These are the local identities of Theorem 3.3. Its rectangular boundary formula does not yet give a formula on a ball.

5.2 Energy on a shrinking ball

Fix a centre $a \in \mathbb{R}^d$ and $R > 0$. At time $t \in [0, R)$ let $r(t) = R - t$. Define the shrinking-ball energy by

$$E_C(t) = \int_{B_{r(t)}(a)} \rho_{\mathrm{en}}(t, x)\, \mathrm{d}^d x. \tag{5.2}$$

On the sphere $\partial B_{r(t)}(a)$, write $n = (x - a)/r(t)$ for the outward spatial unit normal. The normal derivative is $\partial_n u = n \cdot \nabla u$. The tangential gradient is $\nabla_{\tan} u = \nabla u - n \partial_n u$.

Definition 5.2 (Cone energy flux). For a C^2 field, the *outward energy flux through the lateral cone* between times s and t, where $0 \le s < t < R$, is

$$\mathcal{F}_C[s, t] = \int_s^t \int_{\partial B_{R-\tau}(a)} \bigl(\rho_{\mathrm{en}} + \mathbf{f} \cdot n\bigr)\, \mathrm{d}S\, \mathrm{d}\tau. \tag{5.3}$$

Here $\mathrm{d}S$ is spatial sphere area. In $d = 1$, a sphere integral is the sum of its two endpoint values.

The density in (5.3) includes the energy removed as the ball shrinks. It therefore differs from the flux through a fixed spatial boundary. The measure $\mathrm{d}S\, \mathrm{d}\tau$ in this definition is not Euclidean area on the sloping spacetime surface.

Theorem 5.3 (Shrinking-cone energy and local uniqueness). *Let $m \ge 0$ and let u be a real C^2 solution of $(\Box + m^2)u = 0$ on an open neighbourhood of $C^-_{R,a}$. For $0 \le s < t < R$,*

$$E_C(t) + \mathcal{F}_C[s, t] = E_C(s), \qquad E_C(t) \le E_C(s). \tag{5.4}$$

If $u(0, x) = u_t(0, x) = 0$ on $B_R(a)$, then $u = 0$ on the entire closed cone, including its vertex. No assumption on total energy is needed.

Proof. We derive the moving-boundary formula and then use the sign of its integrand.

First, polar coordinates about a express (5.2) as

$$E_C(t) = \int_{S^{d-1}} \int_0^{r(t)} \rho_{\mathrm{en}}(t, a + q\theta) q^{d-1} \, \mathrm{d}q \, \mathrm{d}S_\theta.$$

For $d = 1$, the outer integral sums the directions $\theta = -1$ and $\theta = 1$. The integrand and its time derivative are continuous on each compact truncated cone. Differentiation of this integral and the fundamental theorem of calculus at its upper limit give

$$E_C'(t) = \int_{B_r(a)} \partial_t \rho_{\mathrm{en}} \, \mathrm{d}^d x - \int_{\partial B_r(a)} \rho_{\mathrm{en}} \, \mathrm{d}S, \qquad r = R - t. \tag{5.5}$$

The minus sign comes from $r'(t) = -1$.

Next, the divergence formula for a C^1 vector field on a ball follows from coordinate slices. For $d \geq 2$, hold all coordinates except x^i fixed, writing the remaining vector as y. The endpoints of the slice are $a_i \pm h(y)$, where $h(y) = \sqrt{r^2 - |y - a_{\widehat{i}}|^2}$. Fubini and the one-dimensional fundamental theorem give

$$\int_{B_r(a)} \partial_i f^i \, \mathrm{d}^d x = \int_{|y - a_{\widehat{i}}| < r} \left[f^i(a_i + h(y), y) - f^i(a_i - h(y), y) \right] \mathrm{d}^{d-1} y.$$

On these two sphere graphs, $n_i \, \mathrm{d}S = \pm \mathrm{d}^{d-1} y$. Indeed, their area factor is $r/h(y)$ and their normal component is $\pm h(y)/r$. The equator has sphere area zero. Summing over i proves the ball divergence identity

$$\int_{B_r} \operatorname{div} \mathbf{f} \, \mathrm{d}^d x = \int_{\partial B_r} \mathbf{f} \cdot n \, \mathrm{d}S. \tag{5.6}$$

For $d = 1$ this identity is the endpoint fundamental theorem itself. Substitution of (5.1) into (5.5) now yields

$$E_C'(t) = -\int_{\partial B_{R-t}(a)} \left(\rho_{\mathrm{en}} + \mathbf{f} \cdot n \right) \mathrm{d}S. \tag{5.7}$$

The orthogonal decomposition of ∇u into normal and tangential parts gives the sign:

$$\rho_{\mathrm{en}} + \mathbf{f} \cdot n = \frac{1}{2} \left[(u_t - \partial_n u)^2 + |\nabla_{\tan} u|^2 + m^2 u^2 \right] \geq 0. \tag{5.8}$$

Integrating (5.7) proves (5.4).

Finally, zero initial data make $E_C(0) = 0$. Continuity and positivity force $u_t = 0$ throughout every open shrinking ball. For a point with $|x - a| < R - t$, the whole vertical segment (τ, x), $0 \leq \tau \leq t$, stays inside the cone. Thus $u(t, x) = u(0, x) + \int_0^t u_t(\tau, x) \, \mathrm{d}\tau = 0$. This argument also works for $m = 0$, when energy alone does not detect a constant field. Continuity gives zero on the lateral boundary and at the vertex. ∎

Apply the theorem to the difference of two solutions. Agreement of their initial displacement and velocity on $B_R(a)$ implies agreement throughout the cone. In particular, the data on $\overline{B_T(a)}$ determine the value at (T, a).

> **Corollary 5.4** (Local uniqueness). *Two C^2 Klein–Gordon solutions on a common open time slab $I \times \mathbb{R}^d$ that have the same data on a complete slice $t = t_0$, with $t_0 \in I$, agree throughout the slab. They need not have finite total energy.*

Proof. For any later event, apply Theorem 5.3 to the difference on its backward cone, translating the initial time. Reverse time for earlier events. ∎

5.3 Energy solutions and finite propagation

Lecture 4 constructs solutions in $C(\mathbb{R};H^1)\cap C^1(\mathbb{R};L^2)$. Their spatial gradients are the L^2 Fourier derivatives, and their energy density is defined almost everywhere by (5.1). Values at a single point need not exist. Agreement on a ball therefore means agreement almost everywhere there.

Corollary 5.5 (Energy-class dependence). *For the energy solutions of Theorem 4.3, $E_C(t)\le E_C(s)$ for $0\le s<t<R$. If two such solutions have equal initial data on $B_R(a)$, their values agree almost everywhere on $B_{R-t}(a)$ at each such time.*

Proof. We use the smooth Fourier approximations u_j of Lecture 4, equation (4.6). For every $T>0$, they satisfy

$$u_j\longrightarrow u \text{ in } C([-T,T];H^1),\qquad \partial_t u_j\longrightarrow u_t \text{ in } C([-T,T];L^2).$$

Restriction to any ball has L^2 operator norm at most one. Consequently each integral of the squared terms in the energies at times s and t converges. The smooth inequality passes to the limit. No trace of u_t on the cone boundary is required.

For the difference of solutions with equal local data, its initial energy on $B_R(a)$ is zero. Here its gradient also vanishes there: an H^1 function that is zero on an open ball has zero weak derivatives on that ball, by integration against compactly supported test functions. Thus its velocity vanishes on each shrinking ball at every time. Fix a final ball $B_{R-t}(a)$. All earlier balls contain it. The L^2 fundamental theorem $u(t)=u(0)+\int_0^t u_t(s)\,\mathrm{d}s$, restricted to this fixed ball, proves $u(t)=0$ there, including when $m=0$. ∎

If the initial data vanish outside a closed set K, either version of the dependence result implies

$$\operatorname{supp} u(t,\cdot)\subset\{x:\operatorname{dist}(x,K)\le|t|\}. \tag{5.9}$$

For energy solutions, support means the complement of the largest open set where the function is zero almost everywhere. To prove (5.9) for $t\ge0$, take x with $\operatorname{dist}(x,K)>t$ and choose $t<R<\operatorname{dist}(x,K)$. The base ball $B_R(x)$ has zero data. The solution therefore vanishes on $B_{R-t}(x)$ at time t. Time reversal proves the claim for $t<0$. This is *finite propagation speed*: changing initial data in K can change the field only inside $J^+(K)$ in the future.

5.4 A compact initial velocity pulse

Consider the massless equation in one space dimension, with zero initial displacement and the compact initial velocity

$$u(0,x)=0,\qquad u_t(0,x)=g(x),\qquad g(x)=\begin{cases}(1-x^2)^3, & |x|<1,\\ 0, & |x|\ge1.\end{cases}$$

The function g is C^2, since its endpoint derivatives through order two vanish. Define

$$u(t,x)=\frac12\int_{x-t}^{x+t}g(q)\,\mathrm{d}q. \tag{5.10}$$

The fundamental theorem gives

$$u_t=\frac12[g(x+t)+g(x-t)],\qquad u_x=\frac12[g(x+t)-g(x-t)],\qquad u_{tt}=u_{xx}=\frac12[g'(x+t)-g'(x-t)].$$

These identities verify the equation and both initial data. At (T, a), formula (5.10) uses only $[a - T, a + T]$, exactly the base of its backward cone. For $t \geq 0$, if $|x| > 1 + t$, that interval misses $[-1, 1]$, so $u(t, x) = 0$. This verifies finite speed directly for the pulse.

Inside the cone, the answer can remain nonzero after both endpoints have passed the initial support. For $t > |x| + 1$, the interval contains $[-1, 1]$, and hence

$$u(t,x) = \frac{1}{2}\int_{-1}^{1} (1-q^2)^3 \, \mathrm{d}q = \left[q - q^3 + \frac{3}{5}q^5 - \frac{1}{7}q^7\right]_0^1 = \frac{16}{35}.$$

For example, $u(2, 0) = 16/35$, although both initial data vanish near the endpoints -2 and 2. Finite speed allows dependence on the interior of the base ball. *Huygens' principle* is the stronger assertion that the value at an event depends only on initial data arbitrarily near the base sphere. The massless equation in one space dimension fails that assertion.

5.5 The heat equation has instantaneous tails

Finite speed is a property of the equation. It does not follow merely from solvability of its Cauchy problem. Consider instead $v_t = v_{xx}$ on $\mathbb{R}$, with the same nonnegative compact datum g as initial value. For $t > 0$, define the heat kernel and its solution by

$$K_{\mathrm{heat}}(t,z) = \frac{e^{-z^2/(4t)}}{\sqrt{4\pi t}}, \qquad v(t,x) = \int_{\mathbb{R}} K_{\mathrm{heat}}(t, x-y) g(y) \, \mathrm{d}y.$$

Direct differentiation gives

$$\partial_t K_{\mathrm{heat}} = \left(-\frac{1}{2t} + \frac{z^2}{4t^2}\right) K_{\mathrm{heat}} = \partial_z^2 K_{\mathrm{heat}}.$$

On any compact subset of $t > 0$, derivatives can pass through the integral because g has compact support. Thus $v_t = v_{xx}$. Substitution $y = x - \sqrt{t}\, z$ gives

$$v(t,x) = \int_{\mathbb{R}} \frac{e^{-z^2/4}}{\sqrt{4\pi}} g(x - \sqrt{t}\, z) \, \mathrm{d}z \longrightarrow g(x).$$

The Gaussian has integral one. Boundedness and continuity of g justify the limit by dominated convergence, so this solution has the prescribed data.

For every $t > 0$ and $z \in \mathbb{R}$, the kernel $K_{\mathrm{heat}}(t, z)$ is strictly positive. The datum g is positive on $(-1, 1)$. Their integral is therefore positive for every $x \in \mathbb{R}$, even when $|x| > 1 + t$. The heat equation develops instantaneous tails. Lecture 6 constructs a wave kernel with support confined to the future cone.

Summary. For a real C^2 Klein–Gordon field with $m \geq 0$, energy on a ball shrinking at unit speed decreases by its nonnegative lateral cone flux. Zero local initial data imply zero field throughout the cone, without a global energy assumption. Smooth approximation extends the inequality and local dependence to Lecture 4's energy solutions. Compact initial support expands by at most $|t|$. This conclusion allows interior dependence and does not assert Huygens' principle. The heat equation has no such finite speed bound.

Self-check. At a cone boundary point, let $u_t = 2$, $\partial_n u = 1$, $|\nabla_{\tan} u| = 3$, $m = 2$ and $u = 1$. Compute the cone flux density.

Answer. The density is $\frac{1}{2}[(2-1)^2 + 3^2 + 2^2] = 7$. Omitting the tangential or mass term would give an incorrect smaller flux.

Self-check. For massless $u = C$ with a nonzero constant C, what is its cone energy? Why does this not contradict the zero-data conclusion?

Answer. Every derivative vanishes, so its energy is zero. Its initial displacement is C, whereas the theorem requires zero displacement as well as zero velocity. The vertical integral in the proof retains that displacement.

Self-check. For data supported in $[-1, 1]$, identify the domain of dependence of $(3, 5)$ and decide whether the field can be nonzero there.

Answer. The initial interval is $[2, 8]$. It misses the data support, so the field vanishes at $(3, 5)$ for a C^2 solution. An energy solution vanishes in a neighbourhood of that spatial point at time 3.

Self-check. For the compact velocity pulse, compare $u(2, 0)$ with the data near the base sphere. Which propagation statement does this test?

Answer. The value is $16/35$, while both data vanish near $x = -2$ and $x = 2$. This disproves Huygens' principle in one dimension. It respects finite speed, since the initial support lies inside the base interval $[-2, 2]$.